1-) Project Name

DiaRisk: Diabetes Risk Screening with Machine Learning for SDG 3

2-) Overview

This project aims to build an AI-based tool that estimates a person's risk of diabetes or prediabetes using only simple, self-reported health and lifestyle information, without a blood test. Using the "Diabetes Health Indicators" dataset, we analyse how factors such as BMI, blood pressure, cholesterol, physical activity, age and income are related to diabetes. We then check these self-reported answers against a real blood test (HbA1c) in the NHANES survey to see if they can find people who have diabetes but do not know it. The project is aligned with the following United Nations Sustainable Development Goals:

- **SDG 3 (Good Health and Well-being), Target 3.4:** This target aims to reduce premature deaths from non-communicable diseases such as diabetes through prevention and early treatment.
- **SDG 10 (Reduced Inequalities):** This goal aims to reduce inequalities within countries. Diabetes risk is higher among low-income groups, and a free screening tool that requires no blood test can reach people with limited access to healthcare.

Problem Statement: Diabetes is a growing health problem in Türkiye. According to the IDF Diabetes Atlas, about 16.5% of adults in Türkiye have diabetes and almost half of them (45.5%) do not know it. Diabetes often develops silently and is noticed only after serious complications appear. Prediabetes can often be reversed with lifestyle changes, but only if the person is aware of it. Since diagnosis requires a blood test, many people without symptoms or with limited access to healthcare are never tested.

Analytical Approach: To address this problem, the project starts with a careful analysis of the data before any model is built. We aim to find which health and lifestyle factors are most strongly related to diabetes and how the risk changes across BMI, age and income groups. For example, the analysis investigates whether the risk increases steadily with BMI and age, and whether people with lower income carry a higher risk. It also examines data quality problems such as duplicates, outliers, skewed variables and class imbalance, which directly affect how the model should be trained and evaluated. Finally, we test whether the risk factors stay the same in BRFSS 2019, 2021 and 2023, compare our results with the FINDRISC questionnaire (also used in Türkiye), and check whether only 5–6 questions can give a similar result.

Impact Potential: Our tool will let a user answer a short questionnaire and receive a risk level (low, medium or high) together with the factors that increase their risk most. An LLM will then give general, easy-to-read lifestyle information. This text will not look like medical advice: it will not give a diagnosis, medicine or doses, and it will always advise the user to get a blood test and see a doctor. By encouraging people at risk to change their habits and see a doctor earlier, the tool can support early diagnosis and contribute directly to SDG 3.

3-) Background

Type 2 diabetes is strongly linked to risk factors that can be measured without a laboratory test, such as obesity, high blood pressure, physical inactivity and age. Questionnaire-based risk scores have long been used to decide who should be tested, and machine learning can improve them by learning more complex relationships from large datasets. The "Diabetes Health Indicators" dataset, a cleaned version of the US CDC's 2015 Behavioral Risk Factor Surveillance System (BRFSS) survey, contains 253,680 responses and 21 health and lifestyle features.

Since a risk score alone is often not enough to change behaviour, our project combines a machine learning risk model with an LLM that explains the result in a clear and personal way. Analysing the data is the first step and provides a clean, well-understood foundation for both parts.

A limitation of BRFSS is that diabetes is self-reported, so people who do not know they have diabetes are labelled as healthy. NHANES (National Health and Nutrition Examination Survey) solves this problem because participants answer a questionnaire and also give a blood sample. FINDRISC (Finnish Diabetes Risk Score) is a short questionnaire that has a validated Turkish version, so it is a good benchmark for our tool.

4) Key Objectives / Business Objectives

a) Research Questions:

1. Which self-reported health and lifestyle factors are most strongly related to diabetes and prediabetes?
2. How do BMI, age and income affect the risk of diabetes?
3. How does the strong class imbalance in the data affect the way the model should be built and evaluated?
4. Can simple questions without a blood test give a reliable estimate of diabetes risk?
5. How can the most important risk factors be explained to users in an understandable way?
6. Can self-reported answers find people with undiagnosed diabetes when checked with an HbA1c blood test?
7. Are the risk factors consistent in BRFSS 2015, 2019, 2021 and 2023?
8. How well does FINDRISC work, and can 5–6 questions give the same result?

b) Key Steps:

1. Clean and preprocess the dataset (missing values, duplicates, outliers, data types).
2. Perform exploratory data analysis with descriptive statistics and visualisations.
3. Analyse the relationship of each feature with diabetes using correlation analysis.
4. Train and compare machine learning models to predict diabetes risk, and test them with HbA1c in NHANES.
5. Integrate an LLM that explains the result with general lifestyle information (not medical advice).
6. Test the tool with different user profiles and evaluate its results.

5) Methods and Workflow

a) Datasets:

- **Source:** "Diabetes Health Indicators Dataset" (Kaggle), based on the CDC BRFSS 2015 survey, with 253,680 rows and 22 columns.
- **Scope:** The target variable Diabetes_012 has three classes (0 = no diabetes, 1 = prediabetes, 2 = diabetes). The 21 features include health conditions (high blood pressure, high cholesterol, BMI), lifestyle habits (smoking, physical activity, diet) and demographic information (age, education, income).
- **BRFSS 2019, 2021 and 2023:** The same CDC survey for newer years (243,317, 236,378 and 272,769 rows). The target is diabetes (0/1). The 2023 file has 18 columns.

- **NHANES 2021–2023:** 7,723 US adults (20+); 5,703 have an HbA1c blood test result together with questionnaire answers (diabetes diagnosis, age, sex, BMI, waist, blood pressure, activity, income).
- **Türkiye:** Published results of the TÜİK Türkiye Health Survey (2022 and 2025) and WHO STEPS Türkiye 2017 (obesity, diabetes, blood pressure, smoking, activity; by sex) were collected into one file with 51 rows (turkiye_saglik_ozet.csv), with the source link for each value.

b) Data Cleaning and Preprocessing:

1. **Missing Values and Duplicates:** The dataset has no missing values. 23,899 duplicate rows were removed.
2. **Outlier Removal:** 584 rows with an unrealistic BMI value above 70 were removed. The final dataset has 229,197 rows.
3. **Data Transformation:** All columns were converted from float to integer, and a binary target (0 = no diabetes, 1 = prediabetes or diabetes) was created for the risk analysis.
4. **Skewness Correction:** BMI is right-skewed. A log transformation reduces its skewness from 1.19 to 0.42 (the health-day variables MentHlth and PhysHlth also become less skewed).
5. **New Datasets:** BRFSS 2019, 2021 and 2023 were cleaned with the same rules (duplicates and BMI above 70 removed; final sizes 223,299, 223,119 and 238,825). Because the 2023 file has fewer columns and different income and education codes, years are compared with the 15 common columns. In NHANES, only adults with an HbA1c result were kept, missing activity was set to 0, and HbA1c groups were made with medical cut-offs: 6.5% or more = diabetes, 5.7–6.4% = prediabetes. A person is "undiagnosed" if HbA1c is 6.5% or more but they were never told they have diabetes.

c) Modelling (EDA Phase):

1. **Descriptive Statistics and Class Distribution:** The classes are highly imbalanced: 82.7% no diabetes, 2.0% prediabetes and 15.3% diabetes. BMI and the health-day variables are right-skewed.
2. **Risk Factor Analysis:** The diabetes/prediabetes rate rises from 7.8% at normal weight to 33.5% at BMI 35 or higher, and from 1.8% at ages 18–24 to 25.8% at ages 70–74. It falls from 27.5% in the lowest income group to 11.2% in the highest (Figure 1).
3. **Correlation Analysis:** A Spearman correlation heatmap was created, since most variables are binary or ordinal (Figure 2). General health (0.28), high blood pressure (0.26) and BMI (0.23) are most related to diabetes.
4. **Multicollinearity Control:** No feature pair has a correlation above 0.85 (the highest is 0.44), so no feature needs to be removed.
5. **Time Consistency (BRFSS 2015–2023):** The diabetes rate is stable (15.3%, 15.7%, 14.9%, 16.3%), while obesity slowly rises (36.8% → 40.8%). The correlations of the features with diabetes are almost the same in every year (r = 0.98–0.996), and the same factors (blood pressure, general health, difficulty walking, cholesterol) are on top in every year (Figure 3a, Figure 4).
6. **Self-Report vs Blood Test (NHANES):** 15.8% of people with diabetes (by HbA1c or diagnosis) did not know it, and only 26.3% of people with prediabetes had been told. People with undiagnosed diabetes look like diagnosed patients (mean age 61, BMI 33.7, waist 112 cm), not like healthy people (49, 28.4, 97 cm). Among people not diagnosed, undiagnosed diabetes rises from 0.9% (BMI < 25) to 6.5% (BMI 35+) (Figure 3b, 3c).

7. **FINDRISC and Fewer Questions:** FINDRISC (6 of its 8 questions are available) finds undiagnosed diabetes with an AUC of 0.75. The mean score is 7.7 for normal HbA1c, 11.2 for prediabetes and 12.7 for undiagnosed diabetes (Figure 5a). With a cut-off of 12, it flags 27.4% of people and finds 61.1% of undiagnosed cases. A 3-question score (age, BMI, waist) gives almost the same result (AUC 0.73), so a 5–6 question form that can be used in a pharmacy or by a family doctor is realistic (Figure 3d).

How the Dataset Supports the Project Goals: The EDA shows that simple, self-reported answers that require no blood test carry a clear and medically meaningful signal about diabetes risk, which confirms that a no-lab screening tool is feasible. The strongest factors (general health, blood pressure, BMI and physical activity) are also easy for users to understand and partly change, which makes them suitable for the tool's explanations. NHANES shows that these simple answers can also point to people with undiagnosed diabetes, which we checked with a real blood test.

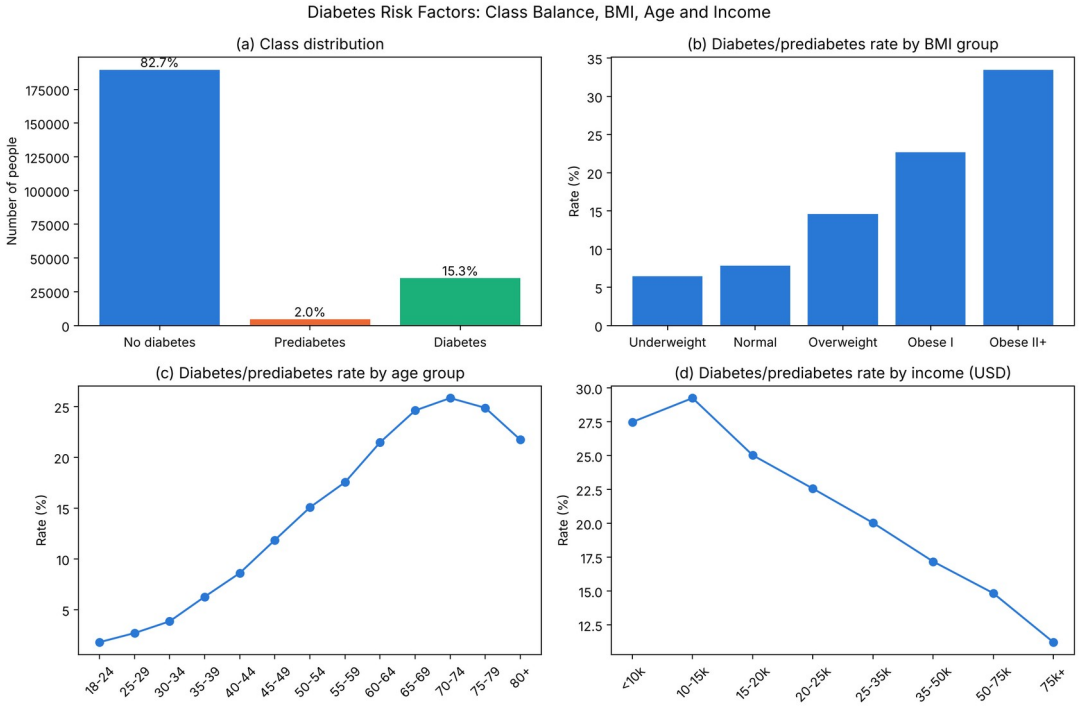


Figure 1. Class distribution and diabetes/prediabetes rate by BMI, age and income

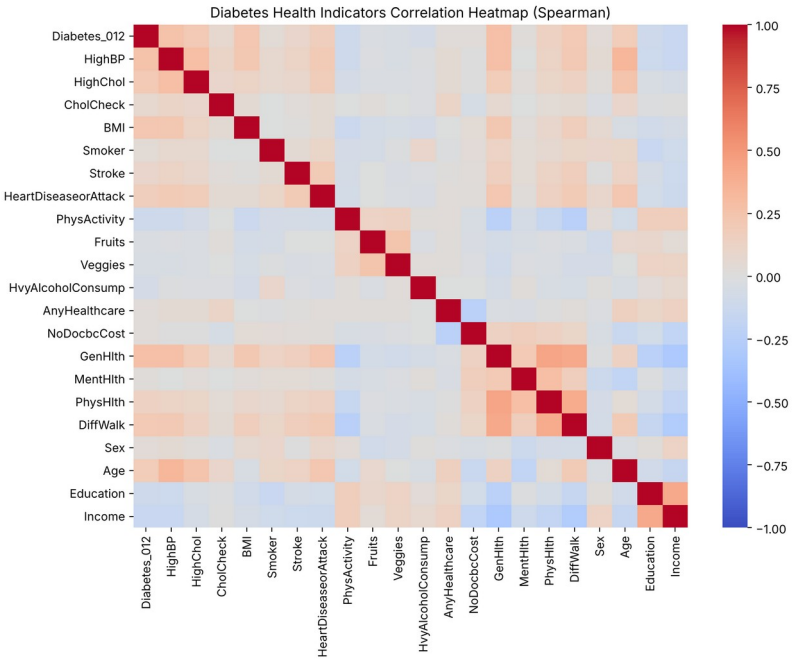


Figure 2. Spearman correlation heatmap of all variables

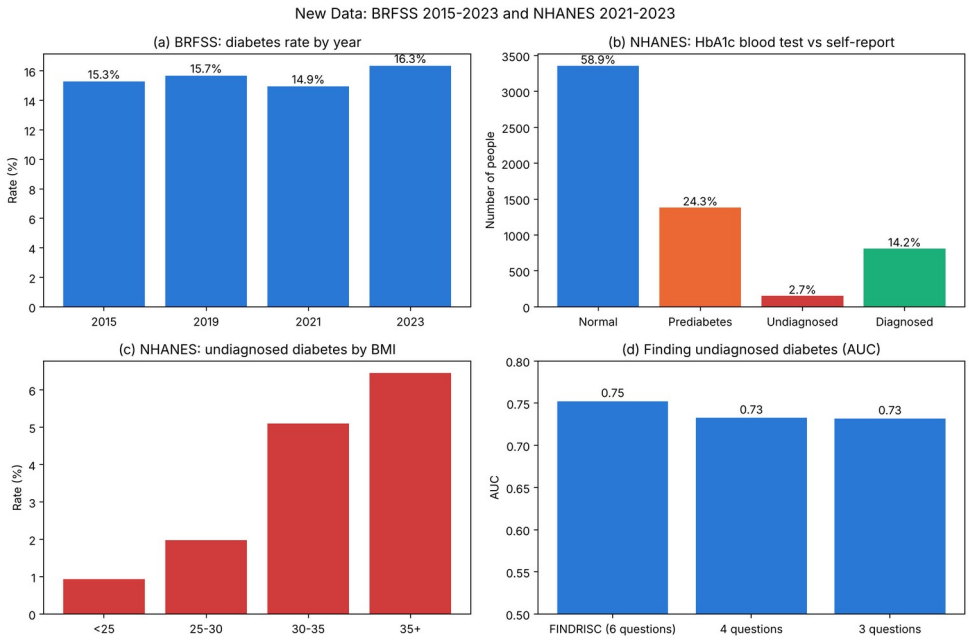


Figure 3. New data: BRFSS diabetes rate by year, NHANES HbA1c groups, undiagnosed diabetes by BMI and AUC of FINDRISC and shorter questionnaires

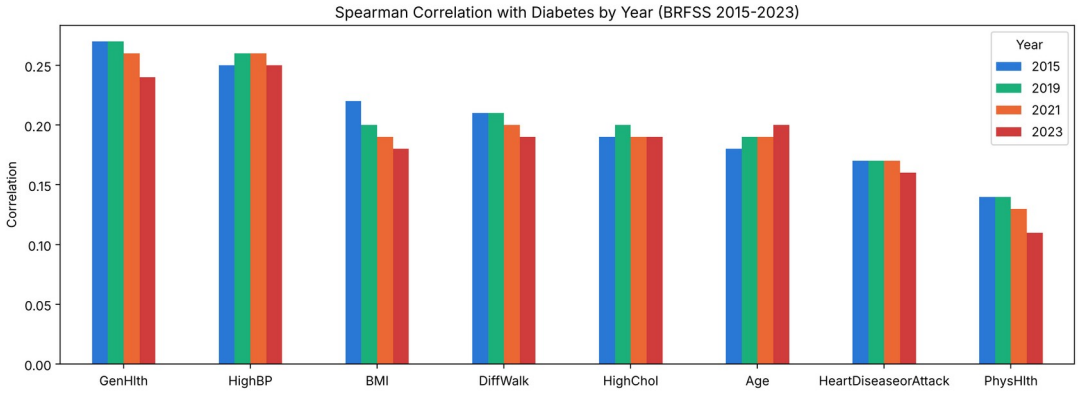


Figure 4. Spearman correlation of the top features with diabetes in BRFSS 2015, 2019, 2021 and 2023

- **Comparison with Türkiye:** In WHO STEPS 2017, diabetes measured with HbA1c (6.5% or more, or medication) is 13.3% in Türkiye, compared with 16.9% in NHANES; obesity is 28.8% vs 40.5% (Figure 5b). TÜİK 2025 values are lower (diabetes 11.9%, high blood pressure 16.9%) because they are self-reported, which again shows that many people do not know their condition. In Türkiye almost half of diabetes cases are undiagnosed (IDF), much more than 15.8% in NHANES, so a short screening questionnaire may be even more useful there.

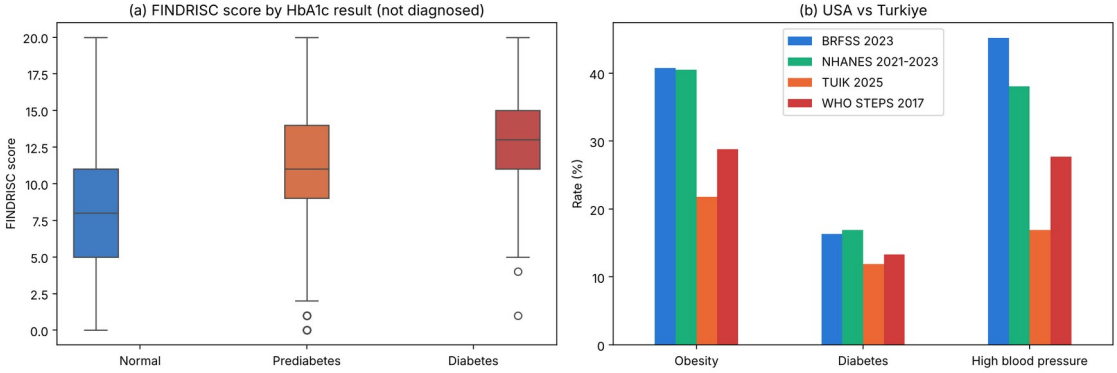


Figure 5. (a) FINDRISC score by HbA1c result, (b) obesity, diabetes and high blood pressure in the USA and Türkiye

Limitations: US data, no survey weights, and FINDRISC is missing 2 of its 8 questions (family history, fruit and vegetables).

d) Deliverables:

1. A Python notebook with the data cleaning, preprocessing and EDA steps (BRFSS 2015–2023 and NHANES).
2. Visual reports including risk factor charts, a correlation heatmap and charts for the new data (BRFSS years, NHANES, FINDRISC, Türkiye).
3. Cleaned datasets ready for machine learning models in the next assignment.

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GitHub: https://github.com/berrin-star/Diabetes-Risk-Screening